

Jonathan J Smith

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Senior Staff Software Engineer | Test Automation Framework Architect

Software Engineer / Test Automation and Framework Architecture with strengths in Java, Python, and Framework Design. Designed and implemented next-generation Python automation framework for video products. Shifted vendor-equivalent testing pre-handoff, reducing third-party certification turnaround by 50 percent, cutting new-vendor test bring-up from weeks. Guided Audio team migration into corporate automation framework. Standardized test patterns across suites, improving reuse by 25 percent and reducing maintenance overhead. Embedded with Headset team to accelerate adoption of a new Python test framework. Led technical evaluation and design discussions.

Key Skills and Expertise

Programming & Scripting: Java • Python

CI/CD & Tooling: CI/CD • Developer Tooling • Test reliability engineering • Versioning automation • Dependency Management • GitHub Actions • Defect isolation • Root cause analysis

Collaboration & Engineering Practices: Technical Leadership • Test planning • Mentoring • Code Review • Process Design

Testing & Validation: Pytest • JUnit 4 • System testing • JUnit 5 • Web UI Automation • Test Lab Automation • ADB • REST/SOAP Integration

Automation & Framework Engineering: Framework Design • Automation Strategy • Automation framework design

Architecture & Design: Architecture • Prototype Design • Infrastructure • Docker

Automation & Framework Architecture: Test data management • Automation Architecture • Capability-Driven Design • UI & Model-Agnostic Abstractions • Reusable components • Playwright

Distributed Systems & Infrastructure: Distributed Execution • Issue triage

Core Engineering Skills: Debugging • AI-augmented test generation

Professional Experience

HP / Poly (formerly Polycom), Austin, TX August 2008 - January 2026

Architect, Python Test Framework (Video) June 2023 - January 2026

Designed and implemented next-generation Python automation framework for video products.

- Shifted vendor-equivalent testing pre-handoff, reducing third-party certification turnaround by 50 percent, cutting new-vendor test bring-up from weeks to days, and reducing reusable test authoring time by 50 percent.
- Set long-term architectural direction and coding standards and mentored a group of five engineers, with two engineers progressing to module-owner responsibilities within two quarters.
- Replaced duplicated test logic across hundreds of tests with shared helper code, reducing test code by 20 percent and enabling faster refactors and rollout of new test capabilities.
- Integrated new web UI automation into an existing containerized GitHub Actions workflow, added npx support to Docker, and shifted web regression execution from manual-only runs toward repeatable automated runs.
- Prioritized high-value acceptance scenarios (including Microsoft certification paths), manually characterized cross-provider controller UI differences, and converted core call workflows into reusable step-based automation with marker/fixture-driven JSON/YAML test data selection.
- Triaged pre-QA failures across automation, firmware, device configuration, and timing/network conditions - most often unannounced UI changes - to isolate root causes early, reduce retest churn, and save the manual QA team two weeks of wasted time per release cycle.

Technical Advisor / SDET, Audio Team May 2021 - June 2023

Guided Audio team migration into corporate automation framework.

- Standardized test patterns across suites, improving reuse by 25 percent and reducing maintenance overhead.
- Improved framework ergonomics with shared abstractions and utilities, increasing reuse by 20 percent.
- Refactored legacy error handling that swallowed I/O exceptions (null/false), enforcing fail-fast behavior at the point of failure and improving API retry/priority handling; improved overall test pass rate by 5 percent or less and reduced random hard-to-reproduce failures by 40 percent.

Technical Advisor / SDET, Headset Team March 2020 - May 2021

Embedded with Headset team to accelerate adoption of a new Python test framework.

- Provided architectural guidance, code reviews, and hands-on development support while mentoring six engineers.
- Enabled distributed engineers to run headset validation remotely through shared lab infrastructure, cutting ad-hoc lab setup time by 80 percent and improving day-to-day execution flexibility.

Technical Lead, Corporate Framework Integration June 2019 - March 2020

Led technical evaluation and design discussions for integrating multiple product groups into a shared corporate automation architecture.

- Delivered a reusable technical evaluation artifact for integrating the Video automation framework into a unified corporate platform, including a first implementation step designed to remain applicable across multiple possible integration directions.
- Decomposed a monolithic automation system into modular components and standardized package and project naming conventions, enabling distributed ownership and reducing build times by 75 percent.
- Architected and drove the Java 8 to 12 migration across 15 projects and 3 product lines, resolving Java 9 module-system breaks, modernizing dependencies, and improving CI and automation stability.
- Implemented semantic versioning automation that, after CI checks passed, automatically generated and released versioned JAR artifacts from standardized commit messages across 15 projects, eliminating manual release tagging.

Lead Technical Designer, Corporate Automation Initiative June 2017 - June 2019

Provided senior-level architectural guidance and technical design leadership for a cross-organization automation initiative.

- Evaluated migration paths for unifying automation across 3 organizations (Video, Audio, Network), improving alignment on a shared framework strategy.
- Built exploratory prototypes and design evaluations that informed cross-team architecture decisions and feasibility tradeoffs, reducing duplicate framework spike efforts by 40 percent across participating teams.
- Conducted framework assessments across 3 product organizations and at least 8 test groups, identifying gaps and unifying database/control patterns; assessments directly informed subsequent team migrations to a unified framework approach, improving migration readiness and driving modularization of REST/third-party control logic into reusable Java libraries.

Lead Framework Designer, Video SQA Automation August 2008 - March 2018

Principal architect and long-term technical steward of the Video SQA automation framework.

- Provided technical leadership for a distributed group of 15 engineers across four countries and five time zones, improving engineering quality and reducing new-test creation time by 30 percent.
- Designed a capability-driven framework architecture with UI-agnostic flows and shared model normalization across SOAP/REST and product variants, improving cross-product portability across hardware generations and 16 endpoint models.
- Architected a four-stage CI quality-gate pipeline — branch merge gates, full suite runs, main branch integration gates, and QA acceptance checks — with stability rules improving deterministic execution across more than 30 releases.

Education

Bachelor of Science (B.S.) Computer Science

Rochester Institute of Technology | Rochester, NY

Coursework emphasis on software engineering and systems programming.

Leadership & Community

Technical Mentor (Informal) | HP / Poly | 2008 - Present

Provided continuous mentoring across teams through the transition from manual QA to automation engineering, including framework thinking, debugging practices, and test design guidance.

CFO & Board Member, Cat Rescue | QA and Test Automation Communities | 2009-Present

Co-manage finances, operations, and long term planning for a 15-year nonprofit rescue, enabling growth from 15-20 cats to a peak of 60.

Phi Kappa Psi Fraternity | Philanthropy Chair

Organized fundraising and community service events.